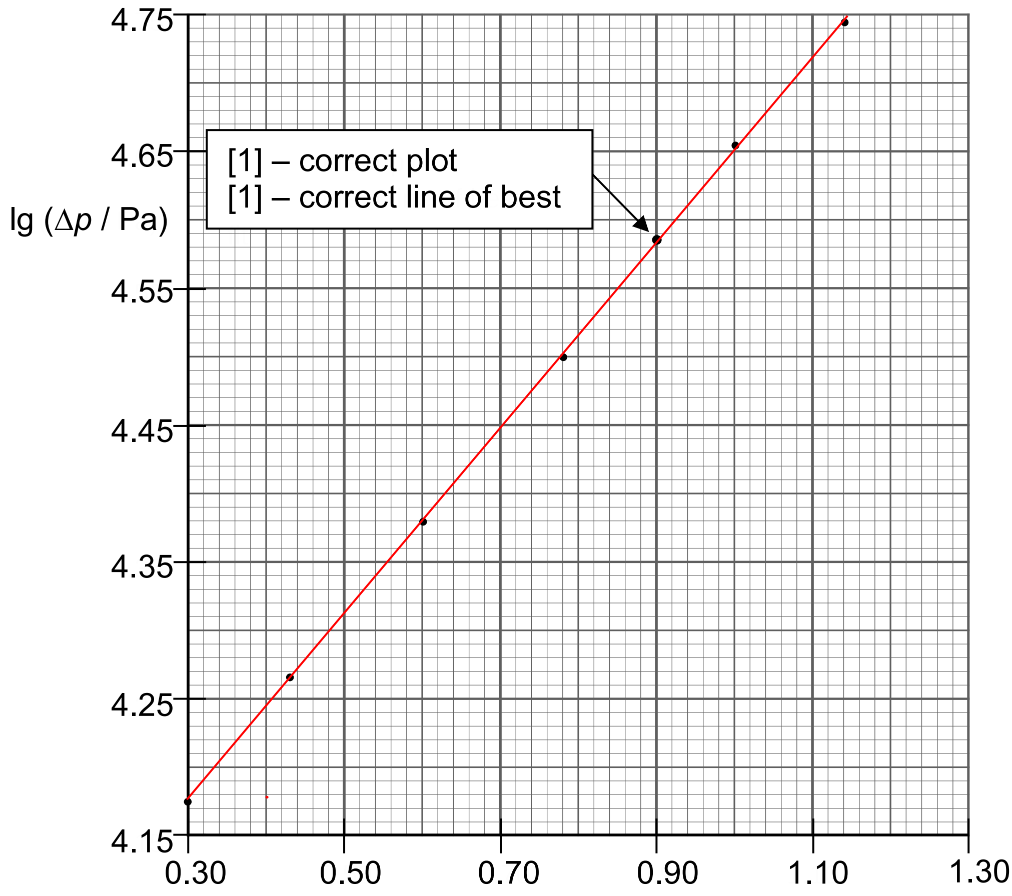
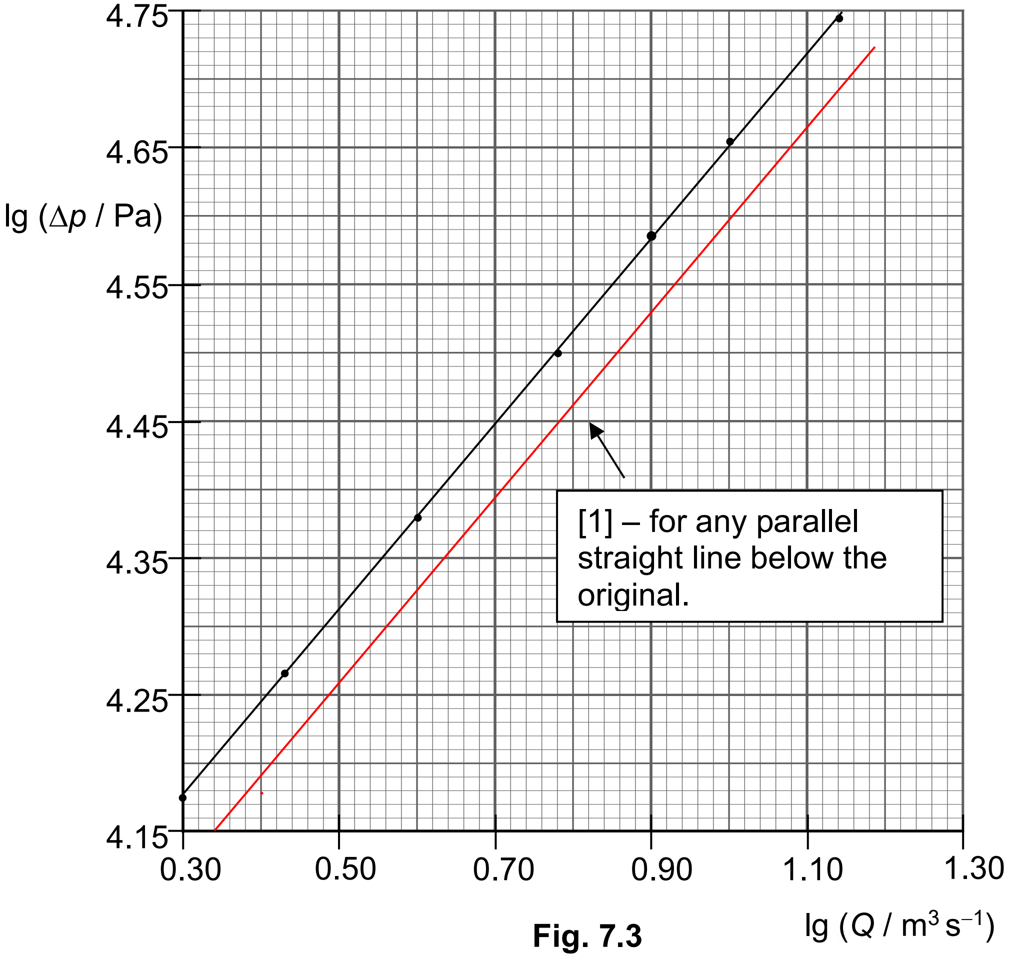
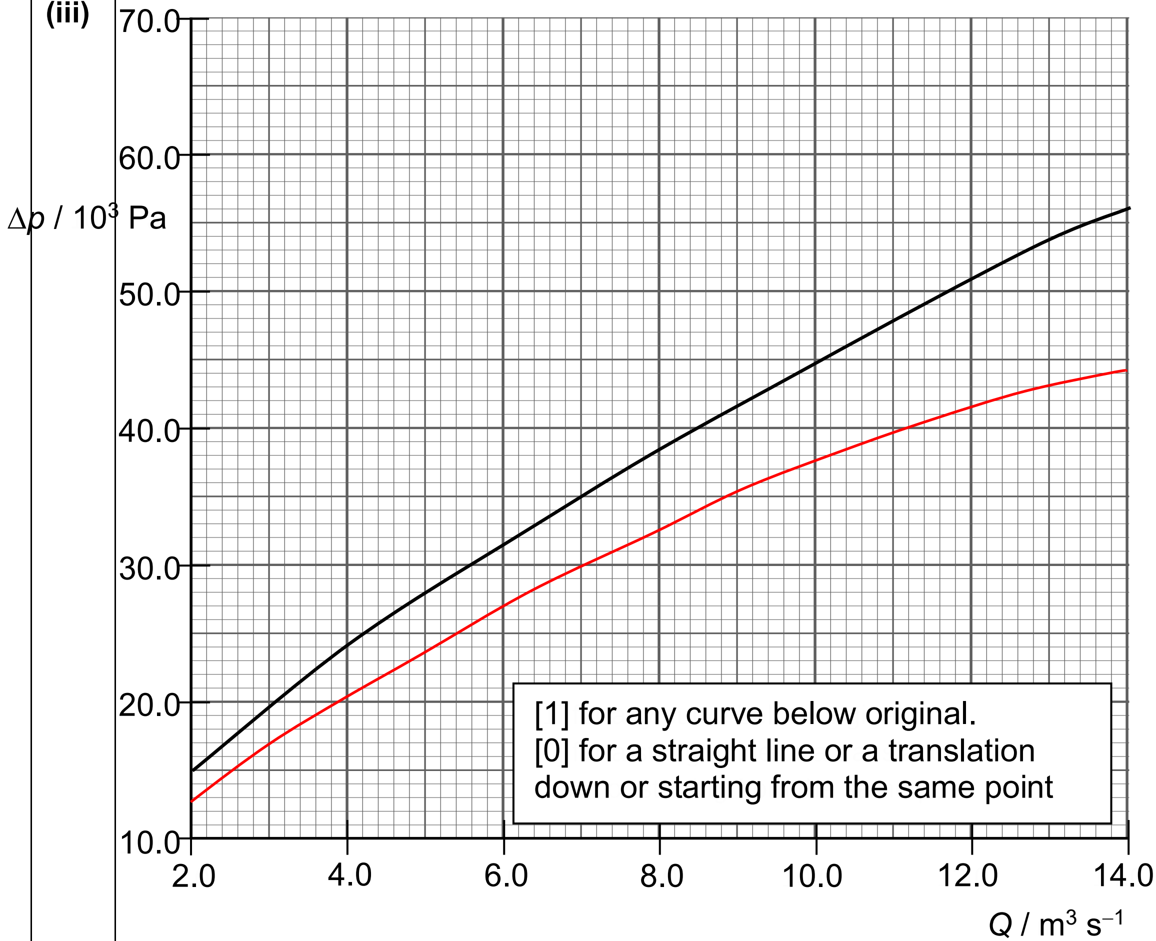
7	(a)	The pressure-drop increases as volume flow rate increases. The rate of increase in pressure-drop decreases as volume flow rate increases.	[1] [1]
	(b)	Due to the orifice plate, the speed of water at upstream position is lower than the speed of water at downstream position. OR The speed of water at downstream position is higher as it passes through a smaller cross section. At lower speed, the pressure becomes higher at the upstream position which results in the pressure difference.	[1] [1]
	(c)	In a vertical position, there is a change in gravitational potential energy between the upstream and downstream positions. This will affect the speed of water, which in turns affect the pressure difference.	[1] [1]
	(d)	(i) At Q = 8.0 m^3 s^{-1}, Δp = 38.5 $\times$ 10^3 Pa (allow 38.0 $\times$ 10^3 Pa) lg Δp = 4.59 (or 4.58) (lg Q = 0.90)	[1] [1]

		(ii)	 **Fig. 7.3**	
		(iii)	Gradient $n = \dfrac{4.650 - 4.300}{1.00 - 0.48}$ (must be > ½ line, correct coord & dp) $= \dfrac{0.350}{0.52}$ $= 0.673$ or 0.67	[1] [1]
		(iv)	Linearising, lg Δp = lg k + n lg Q. Using (1.00,4.650), lg k = 4.650 – 0.673(1.00) = 3.977 $\therefore$ k = 9483 = 9500 or 9480 (allow ecf)	 [1] [1]
		(v)	Unit of Δp $= \dfrac{\text{unit of force}}{\text{unit of area}} = \dfrac{\text{kg m s}^{-2}}{\text{m}^2} = \text{kg m}^{-1}\ \text{s}^{-2}$ Unit of k $= \dfrac{\text{unit of } \Delta p}{\text{unit of } Q^n}$ $= \dfrac{\text{kg m}^{-1}\ \text{s}^{-2}}{\left(\text{m}^3\ \text{s}^{-1}\right)^{0.67}}$ $= \text{kg m}^{-3.0}\ \text{s}^{-1.3}$	[1] [1]

	(e)	(i)	From $k = \dfrac{\rho}{1.2\,A_o^2} \Rightarrow A_o = \sqrt{\dfrac{\rho}{1.2k}} \Rightarrow r^2 = \dfrac{1}{\pi}\sqrt{\dfrac{\rho}{1.2k}}$ $r = \dfrac{1}{\sqrt{\pi}}\left(\dfrac{\rho}{1.2k}\right)^{\frac{1}{4}} = \dfrac{1}{\sqrt{\pi}}\left(\dfrac{1000}{1.2\times 9483}\right)^{\frac{1}{4}}$ $= 0.307$ m (allow ecf)	[M 1] [A 1]
		(ii)	 **Fig. 7.3**	

(iii)



(iv)	Since the density of oil is lower than water, the k value will be lower which leads a lower pressure difference for the same volume flow rate.	[1]
	In order to ensure a more observable height difference in the liquid of the manometer, the diameter of the orifice plate opening should be decreased / the fluid in manometer should be of lower density.	[1]